POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year: 2024

Programme: BCSIT | Full Marks: 100

Course: Mathematics I | Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [10×2=20]

1. Solve the inequality $|2x+3| < 5$
2. Evaluate $\lim_{x \to a} \frac{x^4 - a^4}{x^3 - a^3}$
3. Differentiate the function $f(x) = \ln\sqrt{x^3 + 2x + 3}$
4. Find the marginal profit at the production level 2 if the demand function and cost function of a firm are P=500 - 0.2Q, C=25Q + 1000 respectively.
5. Find the value of the definite integral $\int_1^e \frac{\ln x}{x} dx$.
6. Define singular and non-singular matrix.
7. In how many ways can the letter of the word 'MISSISSIPPI' be arranged taken all at a time.
8. If A = [-3, 2) and B = (1, 4]. Find A∪B and A∩B.
9. Check whether the function $f(x)=x^2-3x+2$ is increasing or decreasing at x=1.
10. If $A = \begin{bmatrix} 2 & 0 & -6 \\ 5 & 1 & 2 \\ 7 & -3 & 0 \end{bmatrix}$, show that the sum of the given matrix and its transpose is the symmetric matrix.

Section "B"

Descriptive Answer Questions

Attempt **any five** questions. [5×10=50]

11. a. In a survey of 260 college students, 64 had taken a mathematics course, 94 had taken a computer science course, 58 had taken a business course, 28 had taken both a mathematics and a business course, 26 had taken both a mathematics and a computer science course, 22 had taken both a computer science and a business course and 14 had taken all three types of courses.
    i. Draw a Venn diagram to present above information.
    ii. How many students were surveyed who had taken none of the three types of courses?

iii. Of the students surveyed, how many had taken only a computer science course?

b. Define tautology and contradiction. If p and q are two statements, then constructing truth table prove that $(p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$ is a tautology.

12. a. Evaluate: $\lim\limits_{x \to a} \frac{\sqrt{3a-x}-\sqrt{x+a}}{4(x-a)}$.

b. $$F(x) = \begin{matrix} 2-x^2 & x<2 \\ 3 & x=2 \\ x-4 & x>2 \end{matrix}$$

Discuss the continuity of the function at x = 2.

13. Find $\frac{dy}{dx}$ (**any two**)

i. $y = \ln(1+sin^2x)$

ii. $x^3 + y^3 = 3axy$

iii. $x = \cos(lnt), \quad y = \ln(sint)$

14. Integrate (**any two**)

i. $\int 2x^3 e^{x^2} dx$

ii. $\int x^n \, lnx \, dx$

iii. $\int_{-\pi/2}^{\pi/2} |sinx| \, dx$

15. a. Show that

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = abc\,(a-b)\,(b-c)\,(c-a).$$

b. If $A = \begin{bmatrix} -1 & 2 & 3 \\ 4 & 0 & -1 \\ -3 & 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 1 & -3 \\ 2 & 0 & -1 \end{bmatrix}$, find $(AB)^T$.

16. a. In how many ways can 4 girls and 4 boys be arranged in a circular table if

i. They may sit anywhere.

ii. They sit alternately.

b. In a group of 6 men and 5 women a committee of 5 have to be formed. In how many ways the committees can be formed if at most 2 women are included?